

GAL Seminar 2

ex 1

Să se calculeze rang A

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 & 2 \\ 2 & -1 & -1 & 0 & 3 \\ 3 & 1 & 0 & 3 & 5 \\ 3 & 3 & 2 & 3 & -1 \\ 5 & 0 & -1 & 3 & 8 \end{pmatrix}$$

ex 2

$$\text{Fie } A = \begin{pmatrix} 1 & 3 \\ 4 & 5 \\ 1 & 1 \end{pmatrix} \text{ și } B = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 0 & 3 \end{pmatrix}$$

$$A \cdot B = ?$$

ex 3

$$\text{Fie } A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 3 \\ 1 & -1 & 1 \end{pmatrix}$$

$$\text{Calculați } A^{-1} = ?$$

ex 4

$$\text{Fie } S \subseteq \{x \in \mathbb{R}^n \mid A \cdot x = 0\}, \quad A \in \mathcal{M}_{m,n}(\mathbb{R})$$

Arătați că $S \subset \mathbb{R}^n$ subspațiu vectorial.

ex 5

Fie un sistem linear omogen și S mulțimea soluțiilor lui.

$$\begin{cases} x_1 - x_2 + 3x_3 + 2x_4 = 0 \\ 3x_1 + 2x_2 + x_3 + 3x_4 = 0 \\ 4x_1 + x_2 + 4x_3 + 5x_4 = 0 \\ 2x_1 + 3x_2 - 2x_3 + x_4 = 0 \end{cases}$$

a) Determinați un sistem de generatori în S

b) Să se det. o mulțime cu 3 elemente care să genereze pe S

c) Fie $v = \left(\frac{7}{5}, -\frac{13}{5}, -2, 1 \right)$

Determinați a, b, c a.î. $v = a \cdot v_1 + b \cdot v_2 + c \cdot v_3$

ex 6

Fie $v_1 = (1 \ 1 \ -1 \ 1)$

$v_2 = (1 \ 0 \ 1 \ 0)$

Verificați dacă vectorii $v = (2 \ 1 \ 0 \ 1)$

și $w = (1 \ 1 \ 1 \ 1) \in \langle \{v_1, v_2\} \rangle = V$

ex 1

Să se calculeze rang A

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 & 2 \\ 2 & -1 & -1 & 0 & 3 \\ 3 & 1 & 0 & 3 & 5 \\ 3 & 3 & 2 & 3 & -1 \\ 5 & 0 & -1 & 3 & 8 \end{pmatrix}$$

Sol:

Obs Când nu avem sistem $\Rightarrow$ în dreapta pivotului se face 0 (la sistem linear nu!)

$$\begin{pmatrix} \boxed{1} & 2 & 1 & 3 & 2 \\ 2 & -1 & -1 & 0 & 3 \\ 3 & 1 & 0 & 3 & 5 \\ 3 & 3 & 2 & 3 & -1 \\ 5 & 0 & -1 & 3 & 8 \end{pmatrix}$$

$$L_2 - 2L_1$$

$$L_3 - 3L_1$$

$$L_4 - 3L_1$$

$$L_5 - 5L_1$$

$$\begin{pmatrix} 1 & \cancel{2}^0 & \cancel{1}^0 & \cancel{3}^0 & \cancel{2}^0 \\ 0 & -5 & -3 & -6 & -1 \\ 0 & -5 & -3 & -6 & -1 \\ 0 & -3 & -1 & -6 & -7 \\ 0 & -10 & -6 & -12 & -2 \end{pmatrix}$$

reînținem coloanele pt
calcul mai
ușor

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \boxed{-1} & -5 & -3 & -6 \\ 0 & -1 & -5 & -3 & -6 \\ 0 & -7 & -3 & -1 & -6 \\ 0 & -2 & -10 & -6 & -12 \end{pmatrix}$$

$$\begin{array}{l} L_3 - L_2 \\ \hline L_4 - 7L_2 \\ L_5 - 2L_2 \end{array}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & \cancel{-5} & \cancel{-3} & \cancel{-6} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 32 & 20 & 36 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \textcircled{1} & 0 & 0 & \textcircled{0} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 \leftarrow 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\text{rang } A = 3$$

minor principal

$$A = \begin{pmatrix} \textcircled{1} & \textcircled{2} & 1 & 3 & \textcircled{2} \\ \textcircled{2} & \textcircled{-1} & -1 & 0 & \textcircled{3} \\ 3 & 1 & 0 & 3 & 5 \\ \textcircled{3} & \textcircled{3} & 2 & 3 & \textcircled{-1} \\ 5 & 0 & -1 & 3 & 8 \end{pmatrix}$$

$$\begin{vmatrix} 1 & 2 & 2 \\ 2 & -1 & 3 \\ 3 & 3 & -1 \end{vmatrix}$$

□

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & \dots & & a_{2n} \\ \dots & & & \\ a_{m1} & & & a_{mn} \end{pmatrix}$$

$$c_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ \dots \\ a_{m1} \end{pmatrix} \quad \dots \quad c_n = \begin{pmatrix} a_{1n} \\ a_{2n} \\ \dots \\ a_{mn} \end{pmatrix}$$

$$A = (c_1 | c_2 | \dots | c_n)$$

$$l_1 = (a_{11}, \dots, a_{1n})$$

...

$$l_m = (a_{m1}, \dots, a_{mn})$$

$$A = \begin{pmatrix} \underline{l_1} \\ \underline{\dots} \\ \underline{l_m} \end{pmatrix}$$

Notation

ex 2

$$\text{Für } A = \begin{pmatrix} 1 & 3 \\ 4 & 5 \\ 1 & 1 \end{pmatrix} \quad , \quad B = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 0 & 3 \end{pmatrix}$$

$$a) \quad A \cdot B = ?$$

$$\begin{aligned} A \cdot B &= \begin{pmatrix} 1 & 3 \\ 4 & 5 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 4 \\ 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1+3 & 2 & 4+9 \\ 4+5 & 8 & 16+15 \\ 1+1 & 2 & 4+3 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 2 & 13 \\ 9 & 8 & 31 \\ 2 & 2 & 7 \end{pmatrix} \end{aligned}$$

Obs data $A \cdot B = (c_1 | c_2 | c_3)$, putem scrie coloanele astea in functie de A ?

$$B = (b_1 | b_2 | b_3)$$

$$c_j = A \cdot b_j$$

Obs

Fie $A \in M_n(\mathbb{R})$ in $\det A \neq 0$

$$A^{-1} = (c_1 | c_2 | \dots | c_n) = ?$$

$$A \cdot A^{-1} = I_n = (e_1 | e_2 | \dots | e_n)$$

$$(A \cdot c_1 | A \cdot c_2 | \dots | A \cdot c_n) = (e_1 | e_2 | \dots | e_n) \Leftrightarrow$$

$$\Leftrightarrow A \cdot c_j = e_j$$

ex 3

$$\text{Fie } A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 3 \\ 1 & -1 & 1 \end{pmatrix}$$

$$\text{Calculati } A^{-1} = ?$$

Sol:

$$\left(\begin{array}{ccc|ccc} \boxed{1} & 1 & 1 & 1 & 0 & 0 \\ 2 & 0 & 3 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} L_2 - 2L_1 \\ \hline L_3 - L_1 \end{array}$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -2 & 1 & -2 & 1 & 0 \\ 0 & -2 & 0 & -1 & 0 & 1 \end{array} \right) \xrightarrow{L_2: (-2)}$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & -2 & 0 & -1 & 0 & 1 \end{array} \right) \xrightarrow{\begin{array}{l} L_1 - L_2 \\ L_3 + 2L_2 \end{array}}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & -1 & 1 & -1 & 1 \end{array} \right) \xrightarrow{L_3 \cdot (-1)}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -1 & 1 & -1 \end{array} \right) \xrightarrow{\begin{array}{l} L_1 - \frac{3}{2} L_3 \\ L_2 + \frac{1}{2} L_3 \end{array}}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{2} & -1 & \frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & -1 & 1 & -1 \end{array} \right)$$

$$\Rightarrow A^{-1} = \begin{pmatrix} \frac{3}{2} & -1 & \frac{3}{2} \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ -1 & 1 & -1 \end{pmatrix}$$

ex 4

Fie $S \subseteq \{x \in \mathbb{R}^n \mid A \cdot x = 0\}$, $A \in M_{m,n}(\mathbb{R})$

Arătați că $S \subseteq \mathbb{R}^n$ subspațiu vectorial.

Sol:

$$\forall x, y \in S, \quad \forall \alpha, \beta \in \mathbb{R}$$

$$\alpha x + \beta y \in S$$

$$\begin{aligned} A(\alpha x + \beta y) &= A(\alpha x) + A(\beta y) \\ &= \alpha \cdot \underset{\substack{= \\ 0}}{Ax} + \beta \cdot \underset{\substack{= \\ 0}}{Ay} = 0 \end{aligned} \quad \square$$

ex 5

Fie un sistem linear omogen și S mulțimea soluțiilor lui.

$$\begin{cases} x_1 - x_2 + 3x_3 + 2x_4 = 0 \\ 3x_1 + 2x_2 + x_3 + 3x_4 = 0 \\ 4x_1 + x_2 + 4x_3 + 5x_4 = 0 \\ 2x_1 + 3x_2 - 2x_3 + x_4 = 0 \end{cases}$$

a) Determinați un sistem de generatori în S

Sol:

Fie V un spațiu vectorial

Vectorii $\{v_1, \dots, v_k\}$ formează un SB dacă

$$\forall v \in V \quad \exists d_1, d_2, \dots, d_k \in \mathbb{R} \quad \text{a.ă.} \quad v = d_1 \cdot v_1 + \dots + d_k \cdot v_k$$

Rezolvăm sistemul :

$$\begin{pmatrix} \boxed{1} & -1 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 4 & 1 & 4 & 5 \\ 2 & 3 & -2 & 1 \end{pmatrix}$$

$$L_2 \leftarrow L_2 - 3L_1$$

→

$$L_3 \leftarrow L_3 - 4L_1$$

$$L_4 \leftarrow L_4 - 2L_1$$

$$\begin{pmatrix} 1 & -1 & 3 & 2 \\ 0 & 5 & -8 & -3 \\ 0 & 5 & -8 & -3 \\ 0 & 5 & -8 & -3 \end{pmatrix}$$

$$L_3 \sim L_2$$

→

$$L_4 \sim L_2$$

$$\begin{pmatrix} 1 & -1 & 3 & 2 \\ 0 & 5 & -8 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$L_2 : 5$$

→

$$\begin{pmatrix} 1 & -1 & 3 & 2 \\ 0 & 1 & -\frac{8}{5} & -\frac{3}{5} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$L_1 \leftarrow L_1 + L_2$$

→

$$\begin{pmatrix} 1 & 0 & \frac{7}{5} & \frac{7}{5} \\ 0 & 1 & -\frac{8}{5} & -\frac{3}{5} \end{pmatrix}$$

→

$$\left(\begin{array}{cc|cc} 1 & 0 & -\frac{7}{5} & -\frac{7}{5} \\ 0 & 1 & \frac{8}{5} & \frac{3}{5} \end{array} \right)$$

$$x_1 = -\frac{7}{5}x_3 - \frac{7}{5}x_4$$

$$x_2 = \frac{8}{5}x_3 + \frac{3}{5}x_4$$

$$\Rightarrow \text{sol} = \left\{ -\frac{7}{5}x_3 - \frac{7}{5}x_4, \frac{8}{5}x_3 + \frac{3}{5}x_4, x_3, x_4 \mid x_3, x_4 \in \mathbb{R} \right\}$$

$$\left(-\frac{7}{5}x_3 - \frac{7}{5}x_4, \frac{8}{5}x_3 + \frac{3}{5}x_4, x_3, x_4 \right)^t =$$

$$= x_3 \left(-\frac{7}{5}, \frac{8}{5}, 1, 0 \right)^t + x_4 \left(-\frac{7}{5}, \frac{3}{5}, 0, 1 \right)^t$$

$$S = \angle \left\{ \left(-\frac{7}{5}, \frac{8}{5}, 1, 0 \right), \left(-\frac{7}{5}, \frac{3}{5}, 0, 1 \right) \right\},$$

b) Să se det. o mulțime cu 3 elemente care
il generează pe S

$$v_3 = v_1 - v_2 = (0, 1, 1, -1)$$

$$v_1 = \left(-\frac{7}{5}, \frac{8}{5}, 1, 0 \right)$$

$$v_2 = \left(-\frac{7}{5}, \frac{3}{5}, 0, 1 \right)$$

$$c) \text{ Fie } v = \left(\frac{7}{5}, -\frac{13}{5}, -2, 1 \right)$$

Determinați a, b, c a.i. $v = a \cdot v_1 + b \cdot v_2 + c \cdot v_3$

sol:

$$\left(\frac{7}{5}, -\frac{13}{5}, -2, 1 \right) = a \cdot \left(-\frac{7}{5}, \frac{8}{5}, 1, 0 \right)$$

$$+ b \cdot \left(-\frac{7}{5}, \frac{3}{5}, 0, 1 \right)$$

$$+ c \cdot (0, 1, 1, -1)$$

$$\left(\begin{array}{ccc|c} -\frac{7}{5} & -\frac{7}{5} & 0 & \frac{7}{5} \\ \frac{8}{5} & \frac{3}{5} & 1 & -\frac{13}{5} \\ 1 & 0 & 1 & -2 \\ 0 & 1 & -1 & 1 \end{array} \right) \quad \begin{array}{l} L_1 : -\frac{7}{5} \\ L_2 : 5 \\ \hline \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & -1 \\ 8 & 3 & 5 & -13 \\ 1 & 0 & 1 & -2 \\ 0 & 1 & -1 & 1 \end{array} \right) \quad \begin{array}{l} L_2 \leftarrow L_2 - 8L_1 \\ \hline L_3 \leftarrow L_3 - L_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & -1 \\ \underline{0} & \underline{-5} & \underline{5} & \underline{-5} \\ \underline{0} & \underline{-1} & \underline{1} & \underline{-1} \\ \underline{0} & \underline{1} & \underline{-1} & \underline{1} \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & -1 \\ \underline{0} & \underline{1} & \underline{-1} & \underline{1} \end{array} \right)$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 0 & -1 \\ 0 & 1 & 1 & 1 \end{array} \right) \quad \begin{array}{l} L_1 \leftarrow L_1 - L_2 \\ \hline \end{array}$$

$$\left(\begin{array}{cc|cc} \overset{a}{1} & \overset{b}{0} & \overset{c}{-1} & -2 \\ 0 & 1 & 1 & 1 \end{array} \right) \quad \Rightarrow \quad \begin{array}{l} a = -c - 2 \\ b = c + 1 \end{array}$$

$$v = (-c - 2) v_1 + (c + 1) v_2 + c \cdot v$$

□

ex 6

Fi $v_1 = (1 \ 1 \ -1 \ 1)$

$v_2 = (1 \ 0 \ 1 \ 0)$

Verificati, dacă vectorii $v = (2 \ 1 \ 0 \ 1)$

$w = (1 \ 1 \ 1 \ 1) \in \langle \{v_1, v_2\} \rangle = V$

Sol:

$v \in V \Leftrightarrow \exists a, b \in \mathbb{R} \text{ a.i. } a \cdot v_1 + b \cdot v_2 \in V$

$\Leftrightarrow$ sistemul
$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \\ -1 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = v$$
 este compatibil $(\text{rang } w)$

$\Leftrightarrow \text{rang} \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ -1 & 1 \\ 1 & 0 \end{pmatrix} = \text{rang} \begin{pmatrix} 1 & 1 & | & 2 \\ 1 & 0 & | & 1 \\ -1 & 1 & | & 0 \\ 1 & 0 & | & 1 \end{pmatrix}$

$\begin{pmatrix} 1 & 1 & | & 2 \\ 1 & 0 & | & 1 \\ -1 & 1 & | & 0 \\ 1 & 0 & | & 1 \end{pmatrix} \xrightarrow[\substack{L_3 + L_1 \\ L_4 - L_1}]{L_2 - L_1} \begin{pmatrix} 1 & 1 & | & 2 \\ 0 & -1 & | & -1 \\ 0 & 2 & | & 2 \\ 0 & -1 & | & -1 \end{pmatrix} \xrightarrow{L_2 \cdot (-1)}$

$\begin{pmatrix} 1 & 1 & | & 2 \\ 0 & 1 & | & 1 \\ 0 & 2 & | & 2 \\ 0 & -1 & | & -1 \end{pmatrix} \xrightarrow[\substack{L_3 - 2L_2 \\ L_4 + L_2}]{L_1 - L_2} \begin{pmatrix} 1 & 0 & | & 1 \\ 0 & 1 & | & 1 \\ 0 & 0 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix}$

$\Rightarrow$ minorul principal al lui $\bar{A} =$ minorul principal al lui $A \Rightarrow v \in V$

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 0 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} L_2 - L_1 \\ \hrightarrow \\ L_3 + L_1 \\ L_4 - L_1 \end{array}$$

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 2 & 2 \\ 0 & -1 & 0 \end{array} \right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & -1 & 0 \end{array} \right)$$

$$\begin{array}{l} L_1 - L_2 \\ \hrightarrow \\ L_3 - 2L_2 \\ L_4 + L_2 \end{array}$$

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{array} \right)$$

$$\text{rang } \bar{A} = 3 \neq \text{rang } A \Rightarrow w \notin V$$